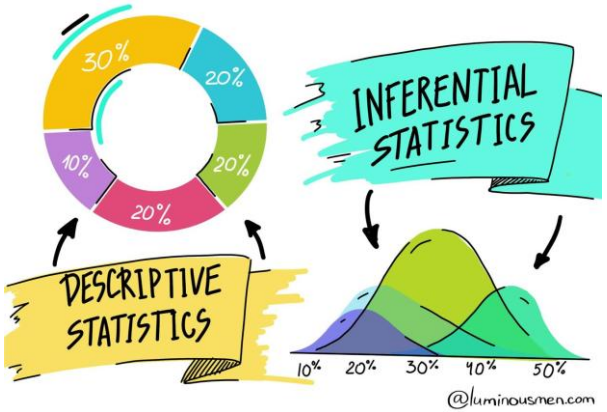
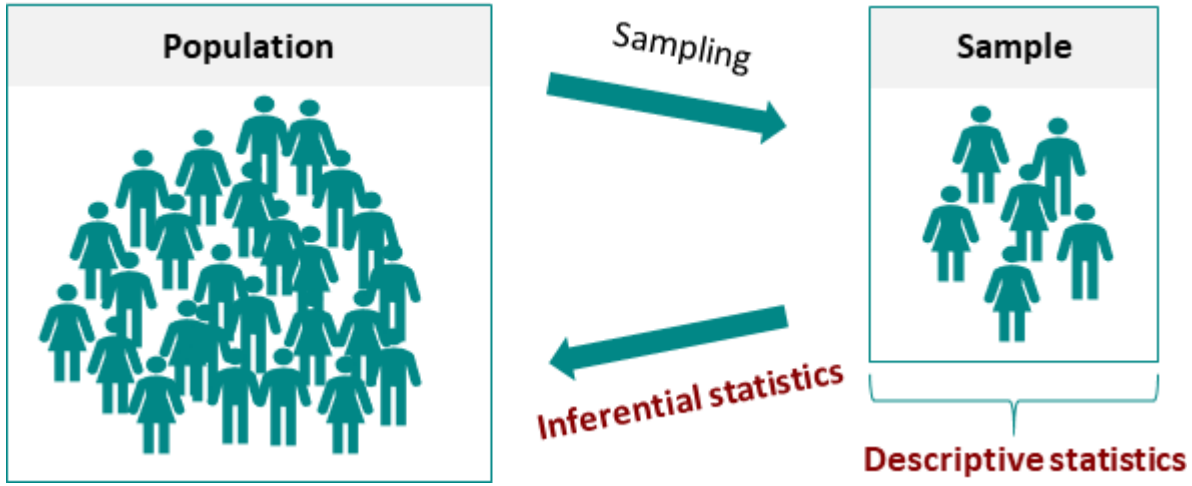
Hypothesis Testing

Why do we need inferential statistics?



Inferential statistics allow us to *infer* or generalize observations made with samples to the larger population from which they were selected.



Descriptive Statistics
- Organise
- Summarise
- Simplify
- Describe and present data

Inferential Statistics
- Generalise from samples to populations
- Hypothesis testing
- Make predictions

What is a Hypothesis?

Research Question
(broad ideas)

→

A specific testable statement (that guides an experiment and statistical analysis)

Research Question – Is online teaching effective?

Hypothesis Statement – Students taught offline perform better than students taught online

(ASSUMPTION) – based on previous studies, observations, experiences, etc.

Null Hypothesis and Alternative Hypothesis

H_0 vs H_1 or H_a

Students taught offline perform better than students taught online

Students taught online perform better than students taught offline

Students taught online vs offline perform equally well on exams (no difference/null)

You perform experiments to check if the Ho holds true or not.
By disproving Ho/null hypothesis, you accept the H_1 or H_A

Do individuals with more years of education have higher income?

Ho – No relationship between years of education and income

H_1 - Individuals with more years of education have higher income

Are leopards stronger than tigers?

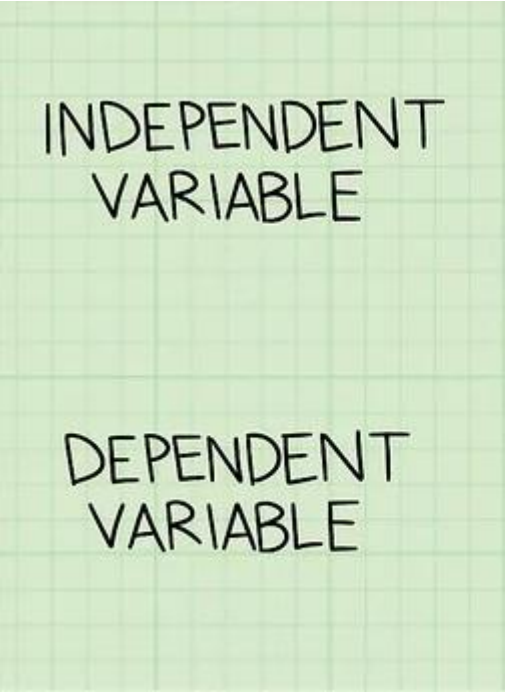
Ho – Leopards and Tigers are equally strong, no difference

H_1 – Tigers are stronger than Leopards

H_2 – Leopards are stronger than Tigers

Variables in a hypothesis

Hypothesis Statement – Students taught offline perform better than students taught online



Variable that is controlled by the experimenter

Teaching method → online / offline

Variable that changes or is affected by the independent variable/s

Exam performance

Does exercise affect anxiety?

H_0 - Exercise has no effect on anxiety

H_1 - Exercise lowers anxiety

H_2 – Exercise increases anxiety

IV – Exercise (exercising, not exercising)

DV – anxiety levels

Are leopards stronger than tigers?

Ho – Leopards and Tigers are equally strong, no difference

H_1 – Tigers are stronger than Leopards

H_2 – Leopards are stronger than Tigers

IV?
DV?

Directionality in a hypothesis

This prediction is typically based on past research, accepted theory, extensive experience, or literature on the topic.

Else your statistical outcome can be misleading, by ignoring other outcomes

Does a technical degree impart technical skills?

vs

High quality of technical education leads to higher technical skills

Ho – Quality of technical education has no effect on technical skills

H_1 - High quality of technical education leads to higher technical skills

Directionality?

Does exercise affect anxiety?

H_0 - Exercise has no effect on anxiety
H_1 - Exercise lowers anxiety
H_2 – Exercise increases anxiety

Directionality?

Does high intensity exercise increase anxiety?

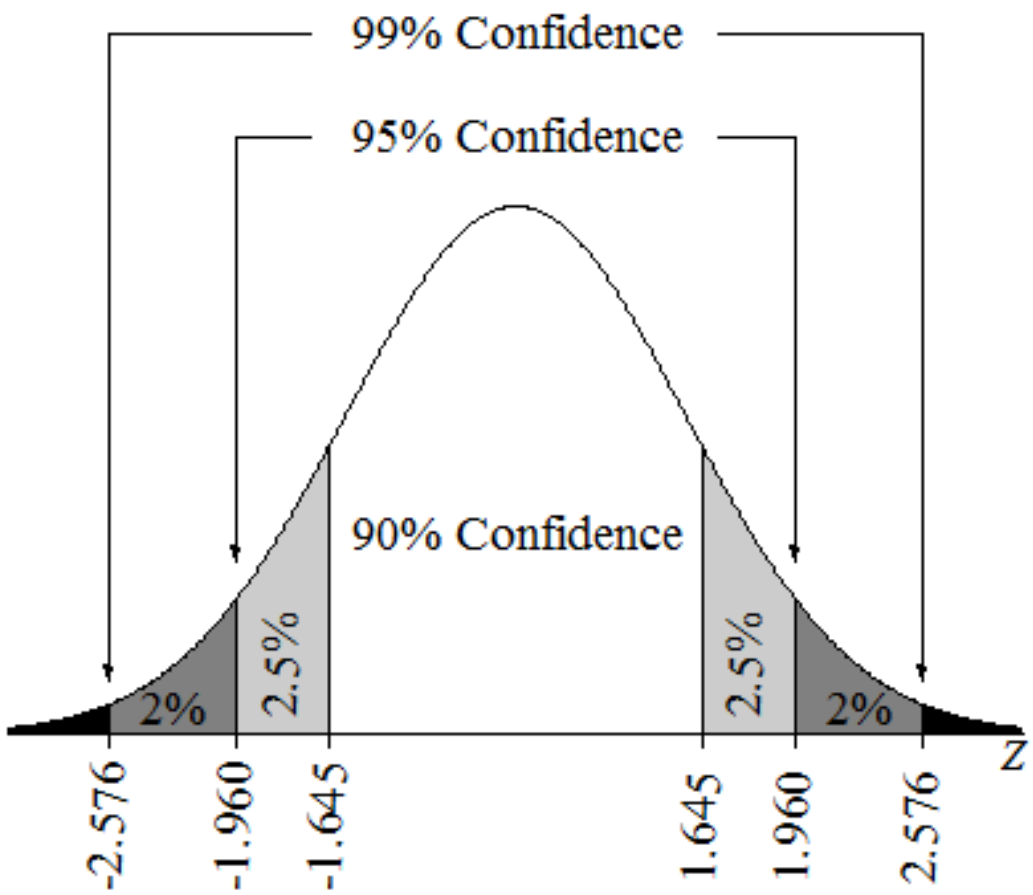
H_0 – High intensity exercise has no effect on anxiety

~~H_1 - High intensity exercise lowers anxiety~~

H_1 – High intensity exercise increases anxiety

You are not interested in the benefits in anxiety, as long as it is not harmful

Confidence Intervals



Confidence Level	α (level of significance)	$Z_{\alpha/2}$
99%	1%	2.575
95%	5%	1.96
90%	10%	1.645

$$CI = \bar{x} \pm z\frac{s}{\sqrt{n}}$$

CI = confidence interval

$\bar{x}$ = sample mean

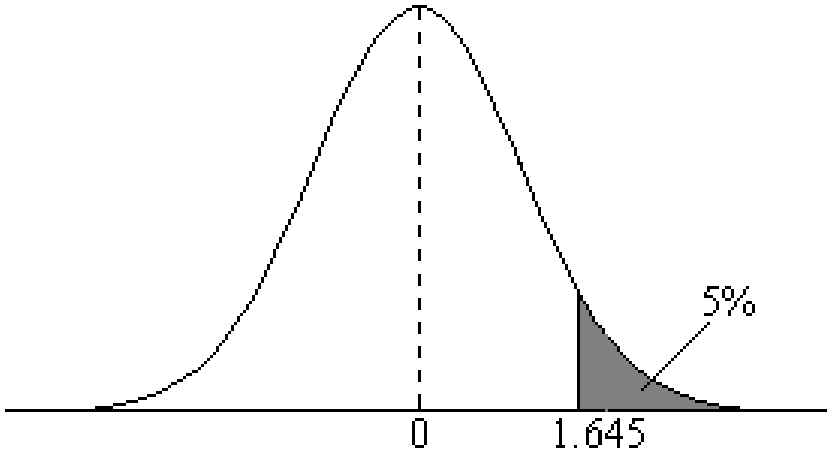
z = confidence level value

s = sample standard deviation

n = sample size

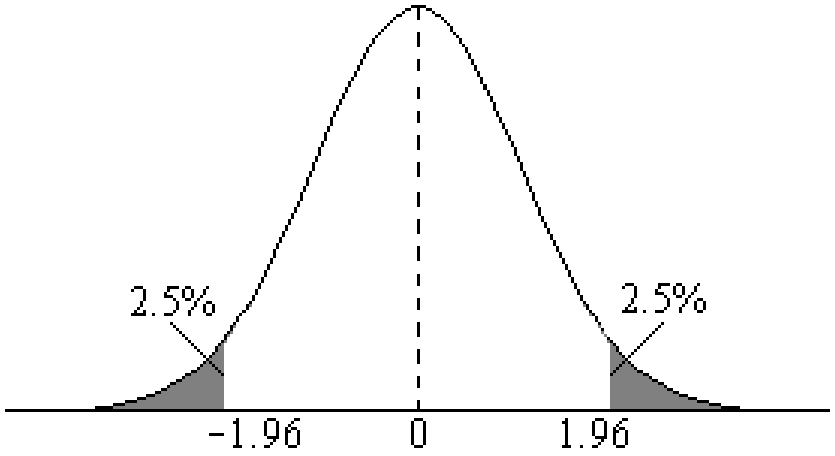
Hypothesis testing

Directional hypothesis



(a) One-tailed test

Non-Directional hypothesis



(b) Two-tailed test

General Rule: Use two-tailed test.

Only if direction is known from prior studies (justified reason), use one-tail test.

Criterion (α) for significance – 5% error (α = 0.05) for most behavioural studies (95 % CI)

If $p \geq 0.05$ → Accept the Ho

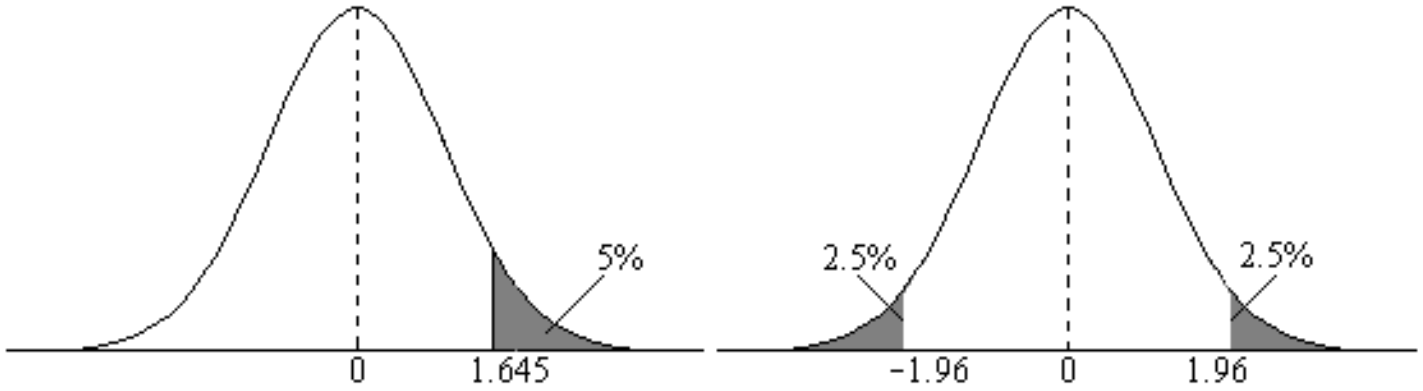
If $p < 0.05$ → Reject the Ho & accept H_A

Exercise effects on anxiety

H_0 - Exercise has no effect on anxiety

H_1 - Exercise lowers anxiety

H_2 – Exercise increases anxiety?

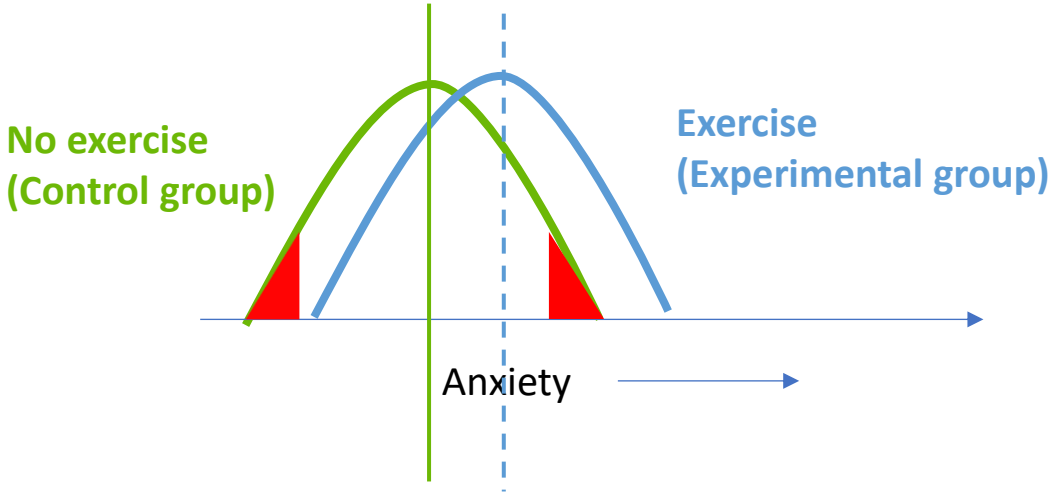


(a) One-tailed test (b) Two-tailed test

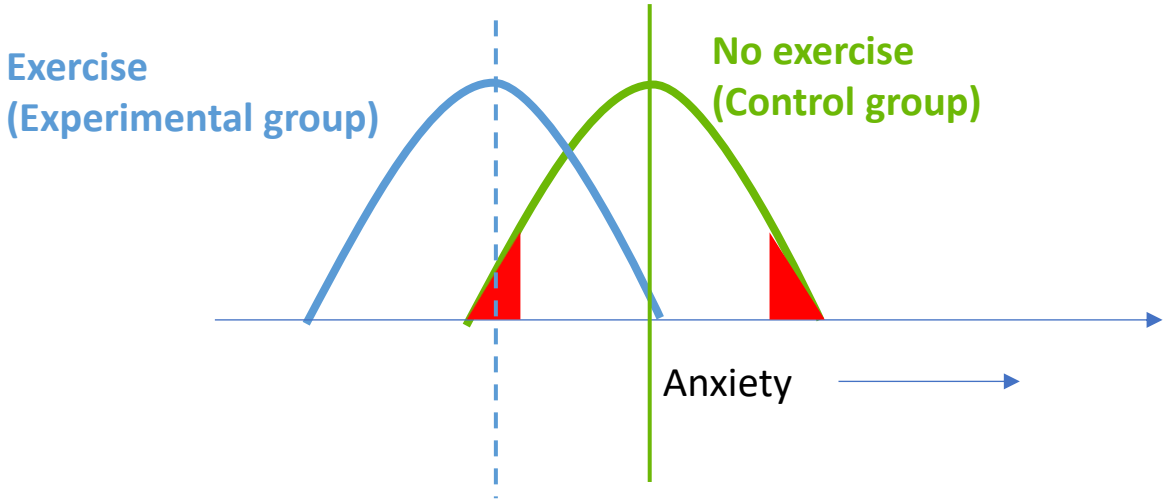
When $p \leq .05$, we reject the null hypothesis - there is a **'significant'** difference between the two groups.

When $p > .05$, we retain the null hypothesis – there is less difference between the groups.

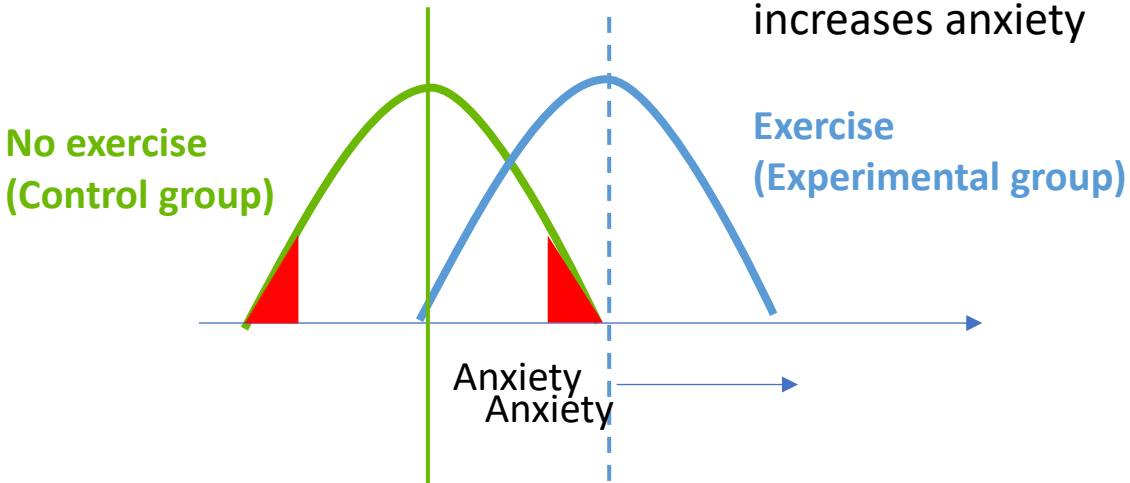
H0 - Exercise has no effect on anxiety



H1 - Exercise lowers anxiety



H2 - Exercise increases anxiety



One-tailed vs two-tailed test

When is a one-tailed test NOT appropriate?

- Choosing a one-tailed test for the sole purpose of attaining significance is not appropriate.

- Choosing a one-tailed test after running a two-tailed test that failed to reject the null hypothesis is not appropriate, no matter how "close" to significant the two-tailed test was.
  - p-hacking

- Using statistical tests inappropriately can lead to invalid results that are not replicable and highly questionable–a steep price to pay to show significance in your results

https://stats.idre.ucla.edu

Another Directional Hypothesis

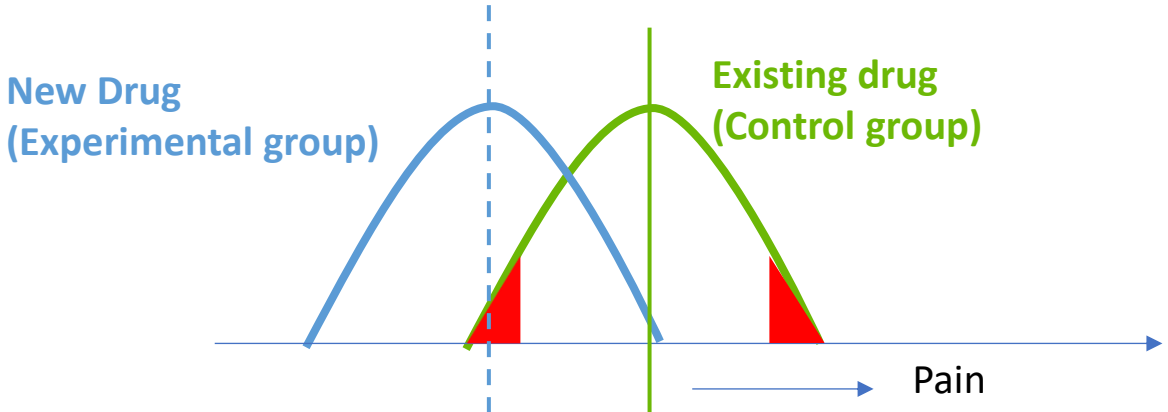
You have a new drug to treat pain that is cheaper than the existing drug and you only want to confirm if the new drug is less effective than the existing drug

Whether the new drug is better than the existing drug does not matter.

H_0 - Null hypothesis – No difference between new drug and existing drug to treat pain

H_1 - Alternate hypothesis – Is the new drug less effective than the existing drug

H_1 – New Drug is less effective



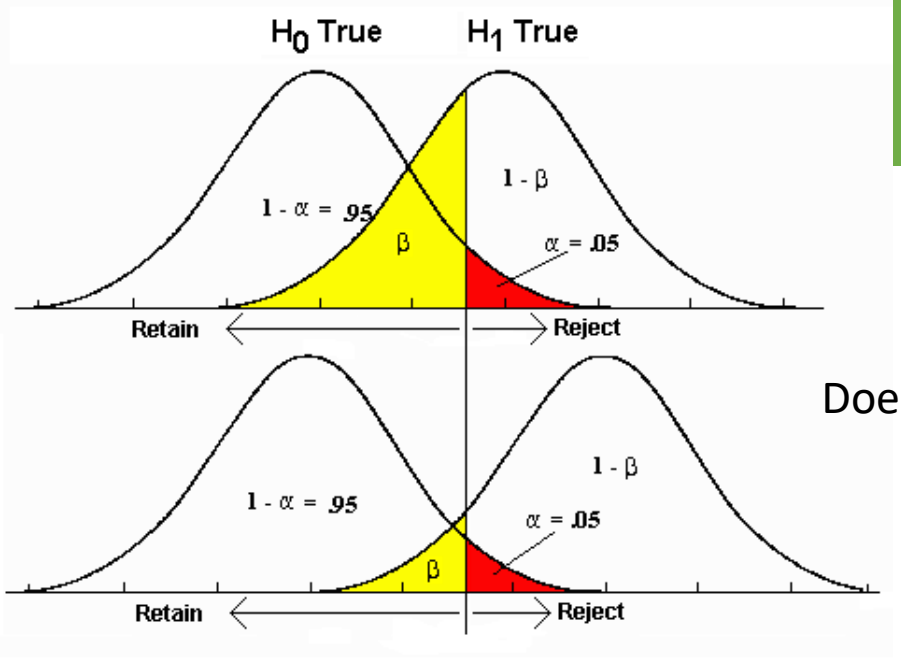
- H_O, H_A, IV, DV, one or two tailed test?

- Smoking is injurious to the lungs
- Videogaming can lower attention span
- Does repetition in advertising improve sales?
- Air pollution is more fatal than COVID19
- Is there a difference in leadership style between men and women?

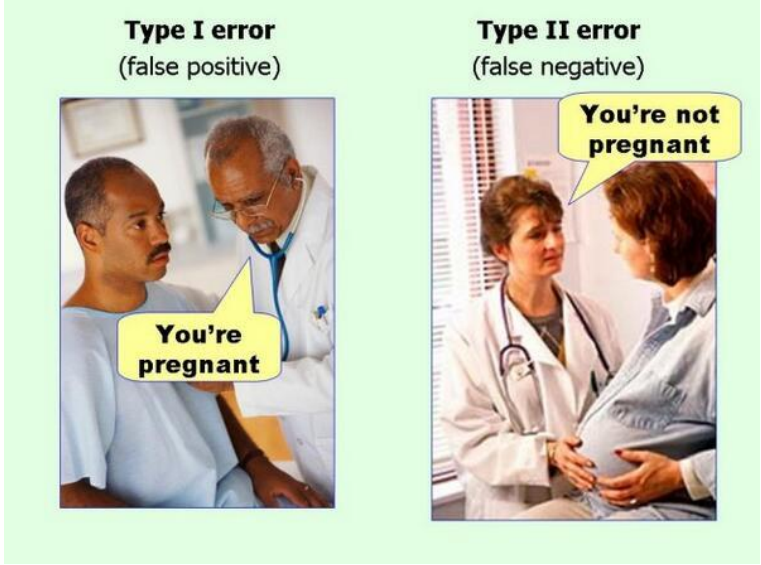
Ever wondered how and why statistics came into picture?

https://nautil.us/how-eugenics-shaped-statistics-238014/

Types of Errors in hypothesis testing



Does repetition in advertising improve sales?



Decision from statistical tests

	Reality: Ho False	Reality: Ho True
Reject Ho	**Correct Decision** Sensitivity/Power 1- β	Type 1 Error "False Positive" α
Accept Ho	Type 2 Error "False Negative" β	**Correct Decision** Specificity 1-α

Observe difference when none exists

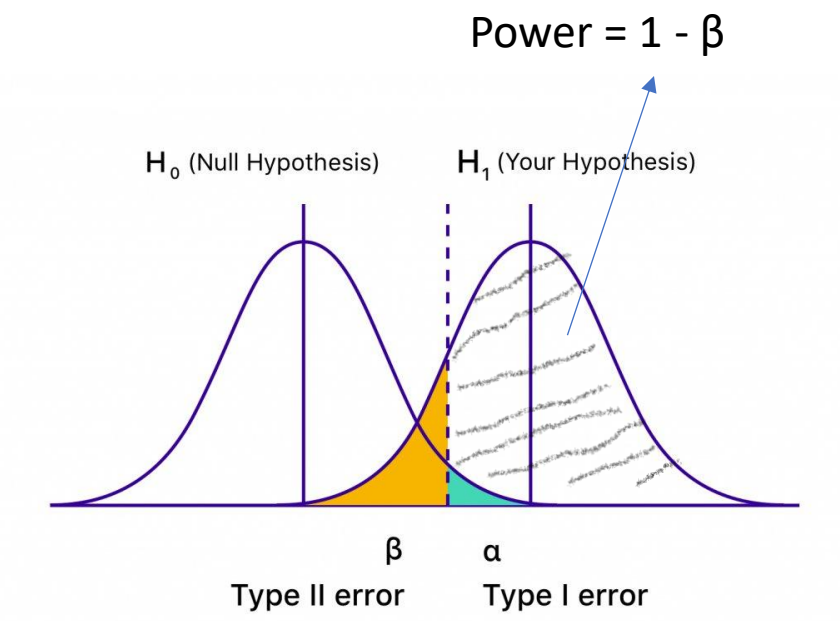
Overreacting!

Fail to find a difference when there is one

Underreacting!

- Sample size it too small (high variability)
- Choosing one-tailed instead of two-tailed test
- Wrong statistical test

Power



- Power - the probability that your test will find a statistically significant differenc when such a difference actually exists.

- Assumption - You will not miss the real effect (Type II error) if you have high power

- It is generally accepted that power should be .8 or greater; that is, you should have an 80% or greater chance of finding a statistically significant difference when there is one.

https://meera.snre.umich.edu/

Power

Power is calculated using statistical software. You need to know –

- What type of test you plan to use (e.g., independent t-test, paired t-test, ANOVA, correlation, regression, etc.)
- The alpha value or significance level you are using (usually 0.05 or 0.01)
- The expected effect size
- The sample size you are planning to use

As your sample size increases, so does the power of your test.

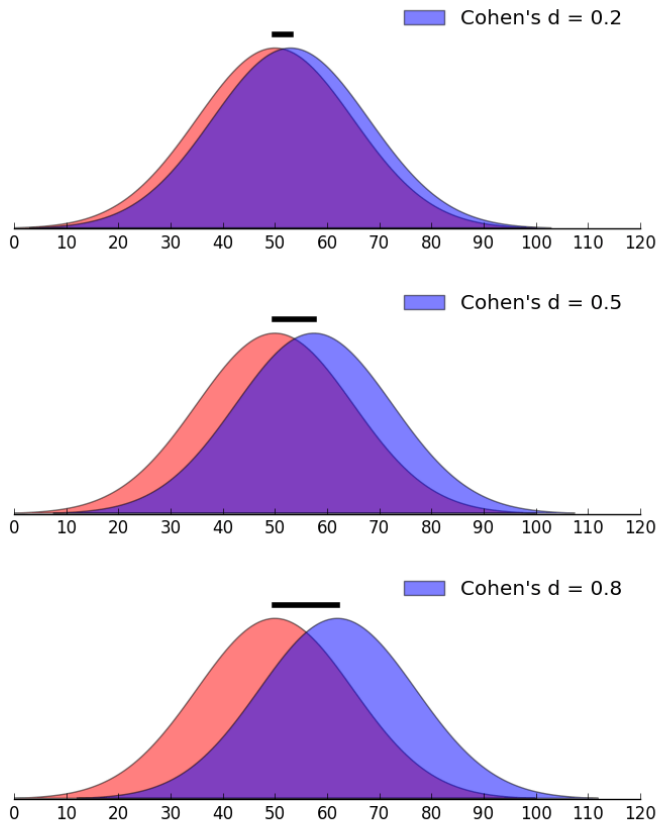
- optimal sample means that you have collected more information -- which makes it easier to correctly reject the null hypothesis when you should.
- A power value is between 0 and 1.
- If the power is less than 0.8, you typically need to increase your sample size.

Effect Size

E.g. you evaluate the effect of a group discussion on student knowledge using pre and post tests on 1000 students. The mean score on the pre test was 83 out of 100 while the mean score on the post test was 84.

- What if you simply found a statistical difference by virtue of a large sample size (> 1000 or 10000)?

- If you calculate the effect size – you get a standard method to defining the importance of the statistical difference



Cohen's d effect size interpretation (for t-test)

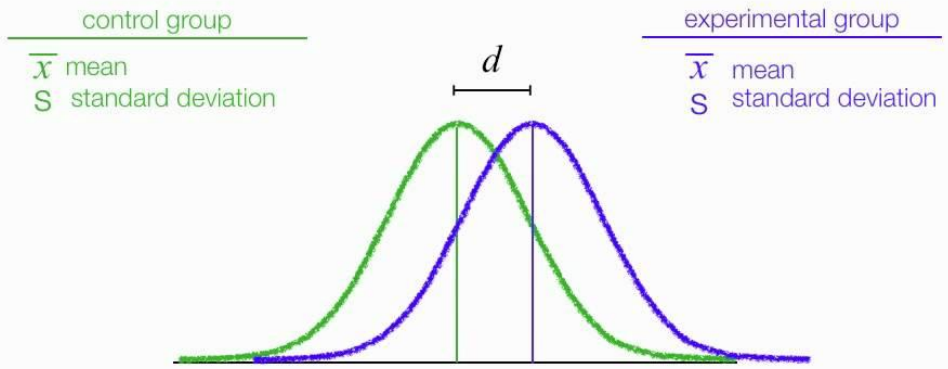
< 0.1 = trivial effect
0.1 - 0.3 = small effect
0.3 - 0.5 = moderate effect
> 0.5 = large difference effect

Effect Size

- Effect size is a quantitative measure of the *strength of a phenomenon*.
- Effect size emphasizes the **size** of the difference or relationship
- Examples:
  - the correlation between two variables (specifically r^2)
    - r=.1 weak, r=.5 moderate, r=.7 strong, r=.9 very strong
  - the regression coefficient in a regression (B_0, B_1, B_2)
    - Relative to model and field
  - the mean differences in t tests (use Cohen's D)
    - d = .2 is small; r = .5 is medium; r = .8 is large
  - The mean differences in ANOVA (use eta)
    - .01 is small, .06 medium, .14 large

$$\text{Cohen's Effect size} = \frac{(\text{Mean}_{\text{treatment}} - \text{Mean}_{\text{control}})}{\text{Standard deviation}_{\text{pooled}}}$$

$$d = \frac{\bar{x} - \bar{x}}{s}$$



Basic formula for sample size - Continuous data

$$\text{Number of samples per group (n)} = \frac{2\,x\,(Z_{(1-\alpha/2)}+Z_{\beta})^2 x\,\sigma^2}{\Delta^2}$$

Where Δ = size of difference, minimal effect of interest

α = significance level (eg 0.05)

β = power, probability of detecting a significant result (typically 80%, 90%)

σ = SD of data

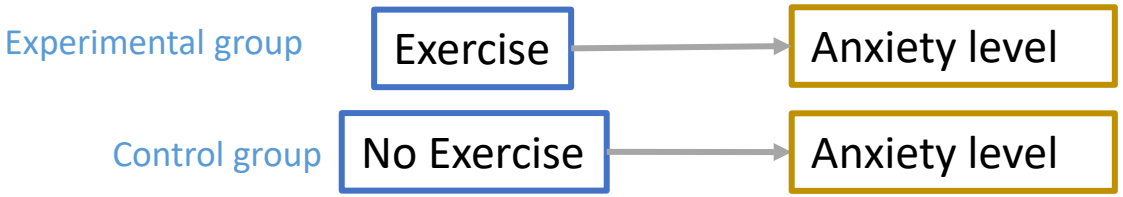
Z_p = points on normal distribution to give required power and significance

DV: Anxiety level

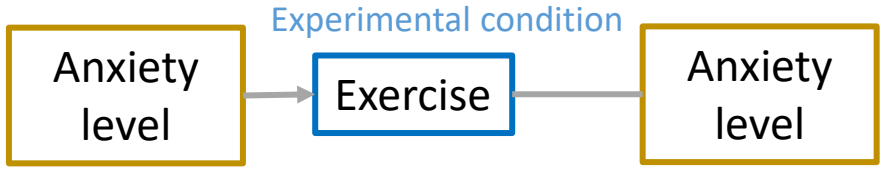
IV: Exercise

Do people who exercise have lower levels of anxiety?

Does exercise lower anxiety?

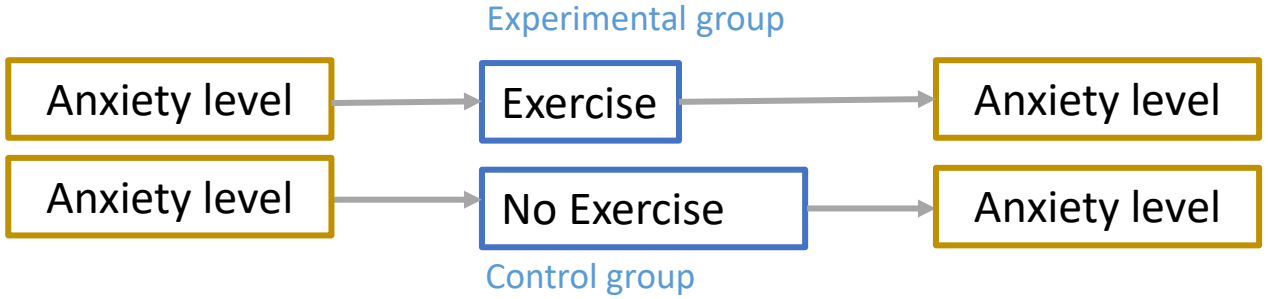


Between groups
(this does not allow you to measure change)



Within group/Repeated measures
(crossover design)

- Participant fatigue
- Longer experimental duration
- Carry over effects



Mixed design
Between groups & Within groups